SAGAR INSTITUTE OF SCIENCE & TECHNOLOGY (SISTec)

DEPARTMENT OF ENGINEERING SCIENCES

MASTER SESSIONAL TEST (MST)

ENGINEERING MATHEMATICS – III (BT-401)

Branch: CSE/AI/IT/ME/CE	Semester: III
Time: 1 Hour 30 Minutes	Maximum Marks: 28

PART – A: Objective Type Questions (18 × 0.5 = 9 Marks)

1. The method based on repeated halving of interval is:

   A) Newton Raphson Method B) Bisection Method C) Gauss Jordan Method D) Simpson’s Rule

2. Newton-Raphson formula is:

   A) $x_{n+1} = x_n - f(x_n)$
   B) $x_{n+1} = x_n + f(x_n)/f'(x_n)$
   C) $x_{n+1} = x_n - f(x_n)/f'(x_n)$
   D) $x_{n+1} = f(x_n)$

3. In Regula-Falsi method, root lies between:

   A) Equal signs B) Opposite signs C) Positive numbers only D) Negative numbers only

4. The forward difference operator is denoted by:

   A) □ B) Δ C) E D) δ

5. The backward difference operator is:

   A) Δ B) E C) □ D) D

6. Shifting operator is represented by:

   A) Δ B) E C) □ D) δ

7. Relation between operators is:

   A) E = 1 + Δ B) E = 1 - Δ C) E = Δ - 1 D) E = □ + 1

8. Interpolation is used to find:

   A) Exact roots B) Missing values C) Derivatives only D) Integrals only

9. Simpson’s 1/3 rule is used for:

   A) Differentiation B) Root finding C) Numerical integration D) Matrix inversion

10. In Trapezoidal rule, graph is approximated by:

    A) Triangle B) Rectangle C) Trapezium D) Circle

11. Numerical integration is also called:

    A) Cubature B) Quadrature C) Factorization D) Iteration

12. Gauss elimination method is used to solve:

A) Differential equations
B) Algebraic equations
C) Simultaneous linear equations
D) Integrals

13. In Gauss Jordan method, matrix is converted into:

A) Diagonal form
B) Triangular form
C) Identity form
D) Symmetric form

14. Jacobi method is:

A) Direct method
B) Iterative method
C) Graphical method
D) Analytical method

15. Crout's method is related to:

A) Interpolation
B) Integration
C) Matrix factorization
D) Differentiation

16. Simpson's 3/8 rule requires intervals to be multiple of:

A) 2
B) 3
C) 4
D) 5

17. The formula for first forward difference is:

A) $f(x + h) + f(x)$
B) $f(x + h) - f(x)$
C) $f(x) - f(x + h)$
D) $hf(x)$

18. The condition necessary for Gauss-Seidel method convergence is:

A) Symmetric matrix
B) Diagonal dominance
C) Zero determinant
D) Unit matrix

PART – B: Short Answer Type Questions

Attempt any TWO from each section.

SECTION – I

19. Using Newton-Raphson method, find the cube root of 2 correct upto 4 decimal places.

20. Find a positive root of $3x = \cos x + 1$ using Bisection Method.

21. Define Forward Difference Operator and Backward Difference Operator.

SECTION – II

22. Using Newton forward interpolation formula, find f(0.18).

23. Using Newton backward interpolation formula, find sin 58°.

24. Evaluate Δe^{ax} and $\Delta^2 e^x$.

SECTION – III

25. Evaluate $\int_0^2 x^3 dx$ using Simpson's 1/3 rule.

26. Solve using Gauss Elimination Method:

$2x + y = 5$
$x + 3y = 6$

27. Apply Gauss Jordan Method to solve:

$2x + y = 4$
$x - y = 1$

PART – C: Long Answer Type Questions

Attempt any ONE.

28.

Find the real root of $x \log_{10} x - 1.2 = 0$ correct upto 5 decimal places using Regula-Falsi Method.

OR

Using Lagrange interpolation formula, find u(6) for the given data:

x	1	2	4	7	8
u(x)	22	30	82	106	206

29.

Evaluate $\int_0^6 dx / (1 + x^2)$ using:

1. Trapezoidal Rule
2. Simpson’s 1/3 Rule
3. Simpson’s 3/8 Rule

OR

Solve by Gauss Seidel Method:

$10x + y + z = 12$

$2x + 10y + z = 13$

$2x + 2y + 10z = 14$